

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. ALL SEM – I Examination, December-2012

ME – 101 Engineering Graphics

Date: 15.12.2012, Saturday

Time: 1:30 p.m to 4:30 p.m.

Maximum Marks: 70

Instructions: (i) Attempt all the questions.

(ii) Figures to the right indicate full marks.

(iii) Make suitable assumptions if required.

SECTION I

Q.1 (a) Choose appropriate answer by rewriting the question

5

- 1 The front view of an elliptical plane may be
a) an ellipse b) a circle c) a straight line d) any of these
- 2 A solid having minimum number of faces is
a) tetrahedron b) triangular prism c) square pyramid d) cube
3. A right circular cone resting on a point of its base circle in the H.P. has the axis inclined at 30° to the H.P. and 45° to the V.P. The angle between the reference line and top view of the axis will be
a) 30° b) between 30° and 45° c) 45° d) more than 45°
- 4 If the distance of a point P from fixed point is more than fixed line then curve formed by point P known as,
a) Ellipse b) hyperbola c) parabola d) eccentricity
- 5 If top view length of line PQ is equal to the true length, then
a) $\beta = \phi$ b) $\beta \neq \phi$ c) $\beta \leq \phi$ d) $\beta \geq \phi$

Q. 1(b) i) Give the application of AutoCAD.

3

- ii) On a map of Gujarat, 1 cm represents 5 kms. Construct a plain scale long enough to measure a distance between Ahmedabad and Baroda. Indicate on it a distance between Ahmedabad and Anand. Distance: (1) Ahmedabad – Baroda – 100 kms. (2) Ahmedabad – Anand – 65 kms. 4
- iii) A Point P is 30 mm below H.P, in third quadrant, and its shortest distance from XY line is 50 mm, Draw its Front View and Top Views. 3

OR

Q.1 (b) A rolling circle of 40 mm diameter is rotate on the directing circle of 120 mm diameter. Point K is on the circumference of rotating circle. Draw the path generate by point K for one complete revolution. Give the name of that path. 10

Q. 2 Line CD is 60 mm long. Point C is in H.P. and behind V.P. Point D is in V.P. and above H.P. line CD is 30° inclined to H.P and 60° inclined to V.P. Draw its projection. 10

OR

Q. 2 Point F is the mid point of 100 mm long line EG. Line EG is 30° inclined to H.P and 45° inclined to V.P. Point F is in H.P. and 15 mm in front of V.P. Draw the projection of line EG. 10

